CS & IT ENGINEERING

Operating System

Virtual Memory

Lecture - 1

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Recap of Previous Lecture

Topic TLB Mapping

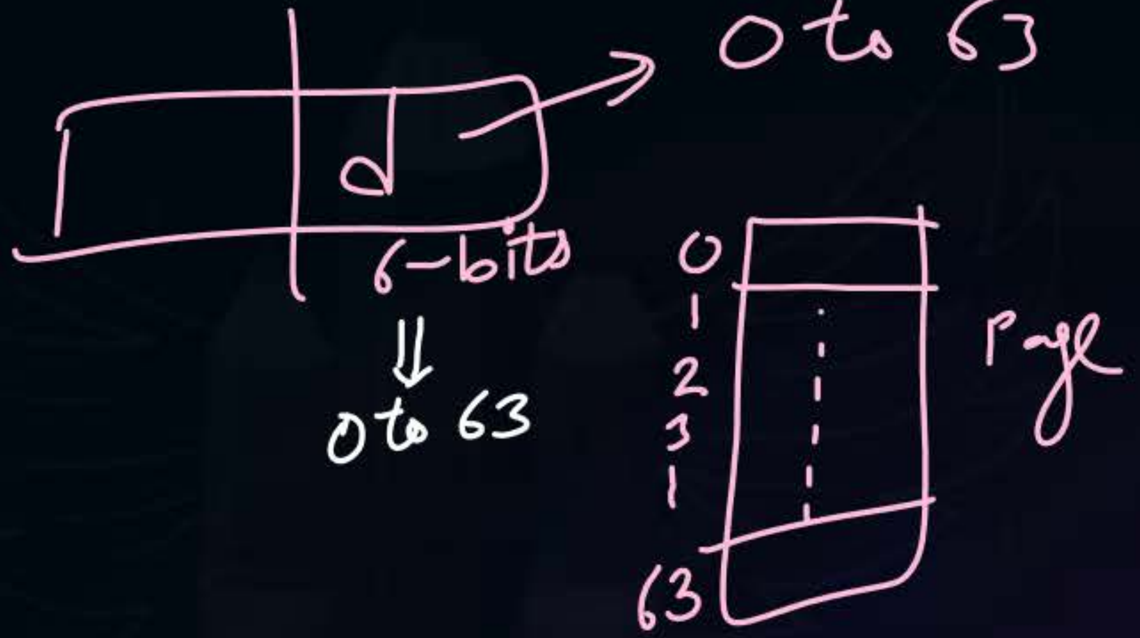
Topic Segmentation

Topics to be Covered

Topic Virtual Memory

Topic Page Fault

page size = 64 B



Topic : Segmentation

- Divide Process in logically related partitions (Segments) Segments are scattered in physical memory
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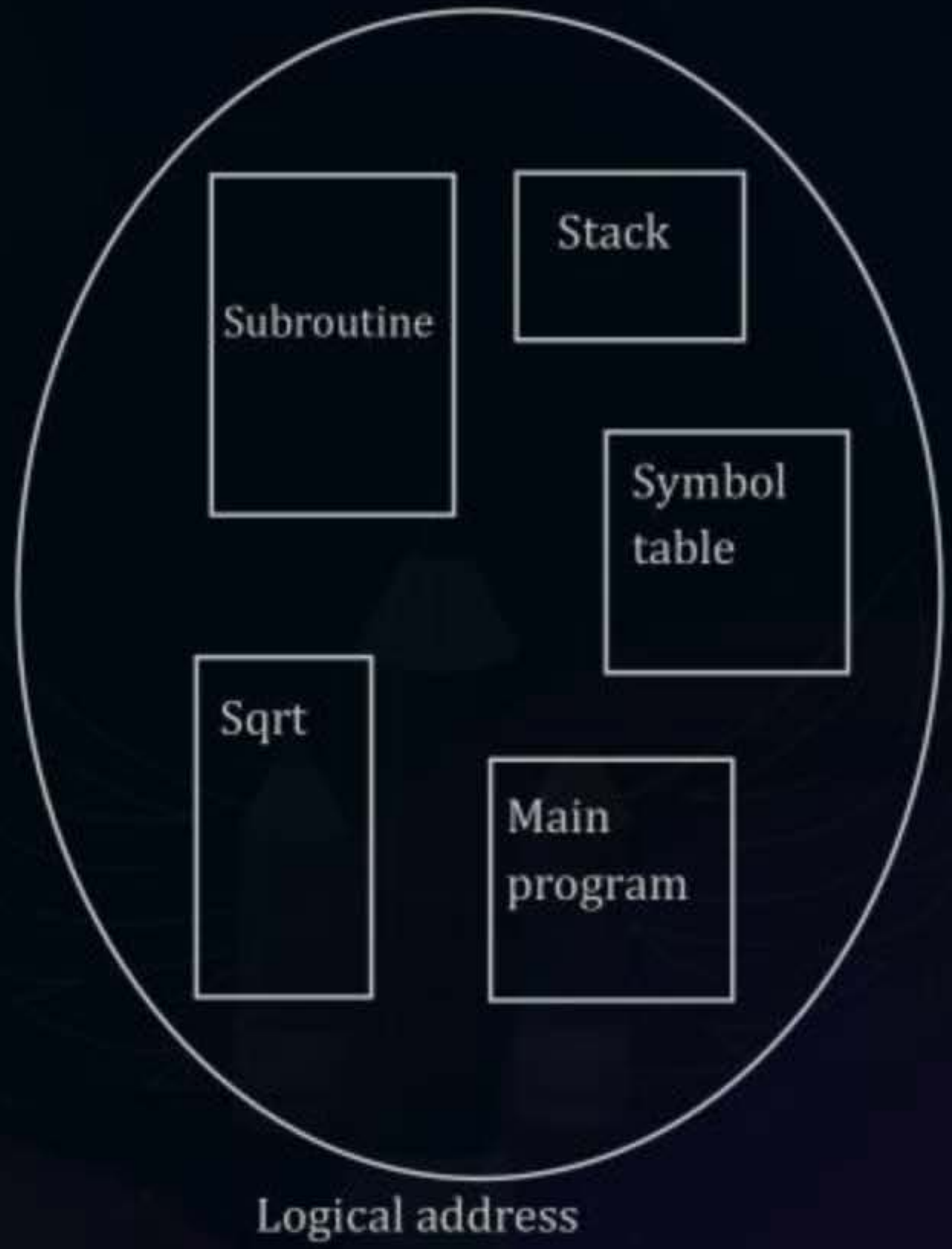
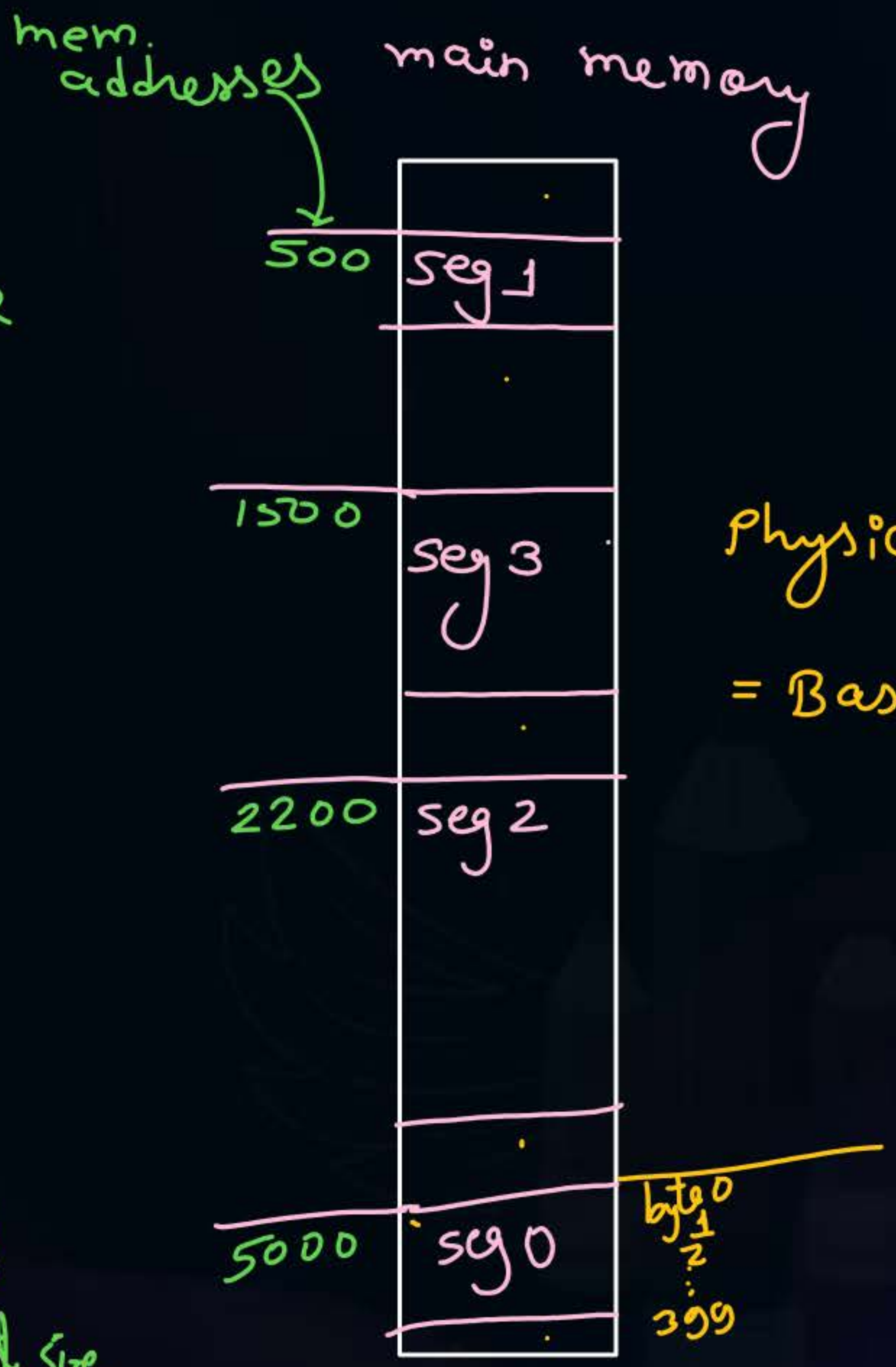


Topic : Segmentation

segment table

0	5000	400
1	500	200
2	2200	1000
3	1500	300

base ↑ limit ↑



Physical add.

= Base + d

Logical address

S	d

→ based on largest segment size

Base :- starting add. of seg. in memory

limit :- no. of bytes in the segments

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if logical address

s	d
1	300

check if $d < \text{limit}$

⇓

yes ⇒ base + d → physical add.

[NAT]

#Q. Find physical address for the following requests?

S	d	Physical Address
1	165	500+ 165 = 665
2	1059	fault
0	400	fault
3	249	1500 + 249 = 1749

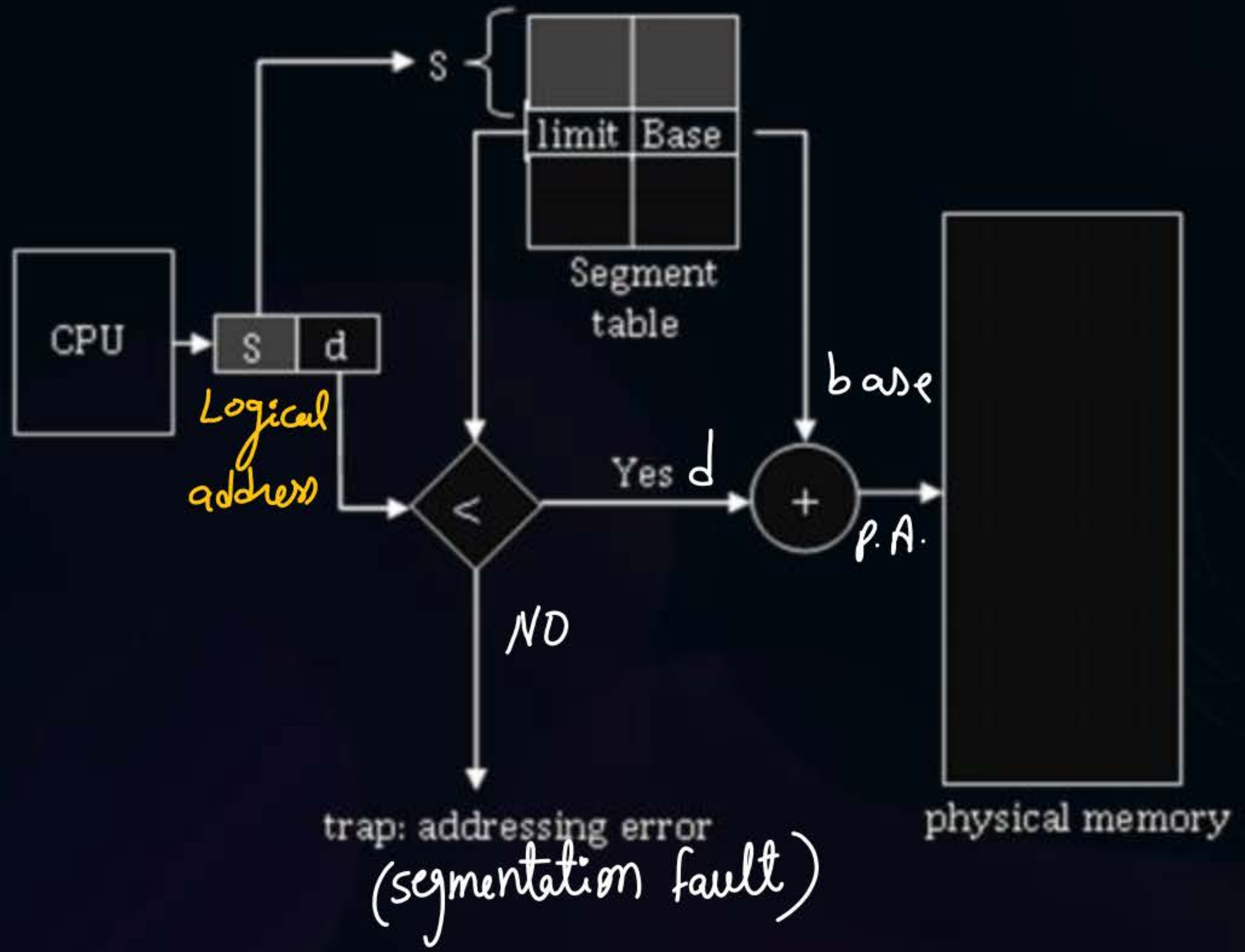
segment table

	base	limit
0	5000	400
1	500	200
2	2200	1000
3	1500	300

Topic : Segmentation

- Size of segment can vary, so along with base, keep limit information also
- Limit defines max number of ~~words~~ bytes within the segment

Topic : Segmentation

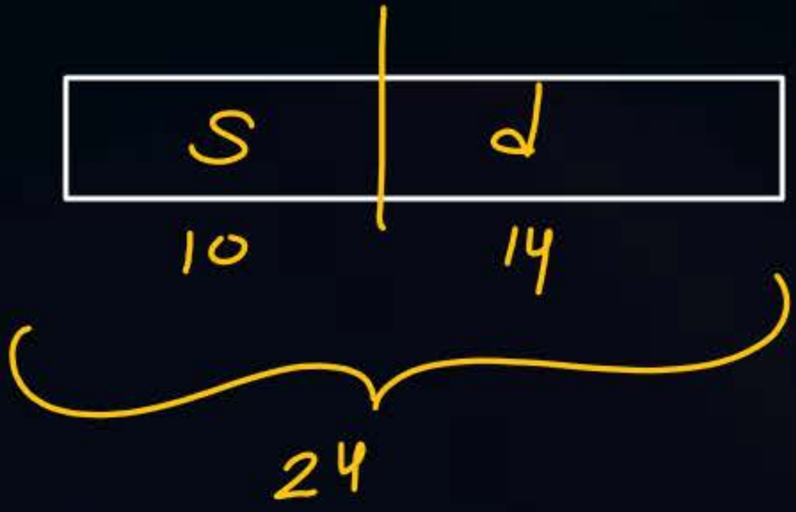


[NAT]

#Q. Maximum segment size = 16KB $\rightarrow = 2^{14}$ B

Number of segments in process = 2^{10}

Logical address = __24__ bits ??



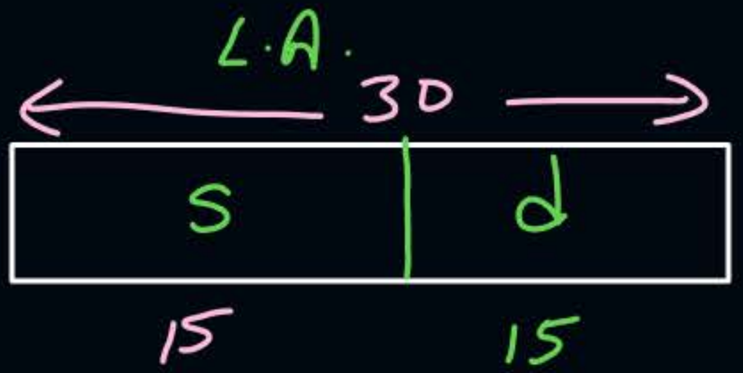
Ques) no. of segments = ______ ?

L.A. = 30 bits

segment sizes between 2kB to 32kB

2^{15} B

Soln



based on largest segment

$\Downarrow$

no. of segments = 2^{15} Ans.

Topic : Segmentation

- Segmentation suffers from external fragmentation

Memory mgmt techniques

- Contigous
  - fixed partition $\Downarrow$ Internal fragmentation
  - variable partition $\Downarrow$ external fragmentat^n
- non-contiguous
  - Paging $\Downarrow$ Internal fragmentat^n
  - segmentation $\Downarrow$ External fragmentat^n

Topic : Virtual Memory

- Feature of OS
- Enables to run larger process with smaller available memory

Topic : Virtual Memory

Process

Page table

	Process		Page table
	Page 0	000	4
	Page 1	001	
	Page 2	010	3
	Page 3	011	1
	Page 4	100	0
	Page 5	101	
	Page 6	110	~~3~~
	Page 7	111	

Physical memory

0	00	Page 4
1	01	Page 3
2	10	Page 0
3	11	Page ~~6~~ 2

Hard disk

Page 1

Page 2

Page 5

Page 7

Page6

p	d
2	. . . .

CPU generates L.A.

↓

search in P.T.

⇓

page 2 not available in mm

↓

Page fault

↓

OS services page fault

↓

OS brings faulted page in mm → Page table is updated → instruction which caused page fault will restart

(if needed a page replaced from mm)

Topic : Demand Paging

- **Demand Paging:**

  Bring pages in memory when CPU demands

- **Page Fault:**

  When the demanded page is not available in physical memory

Demand Paging

normal demand paging

when a process arrives, a few of the pages will be brought into mm & remaining on demand.

Pure demand paging

no any page of process brought to mm at arrival

Topic : How to Check Page Hit/Fault

valid invalid bit

Page table

	f

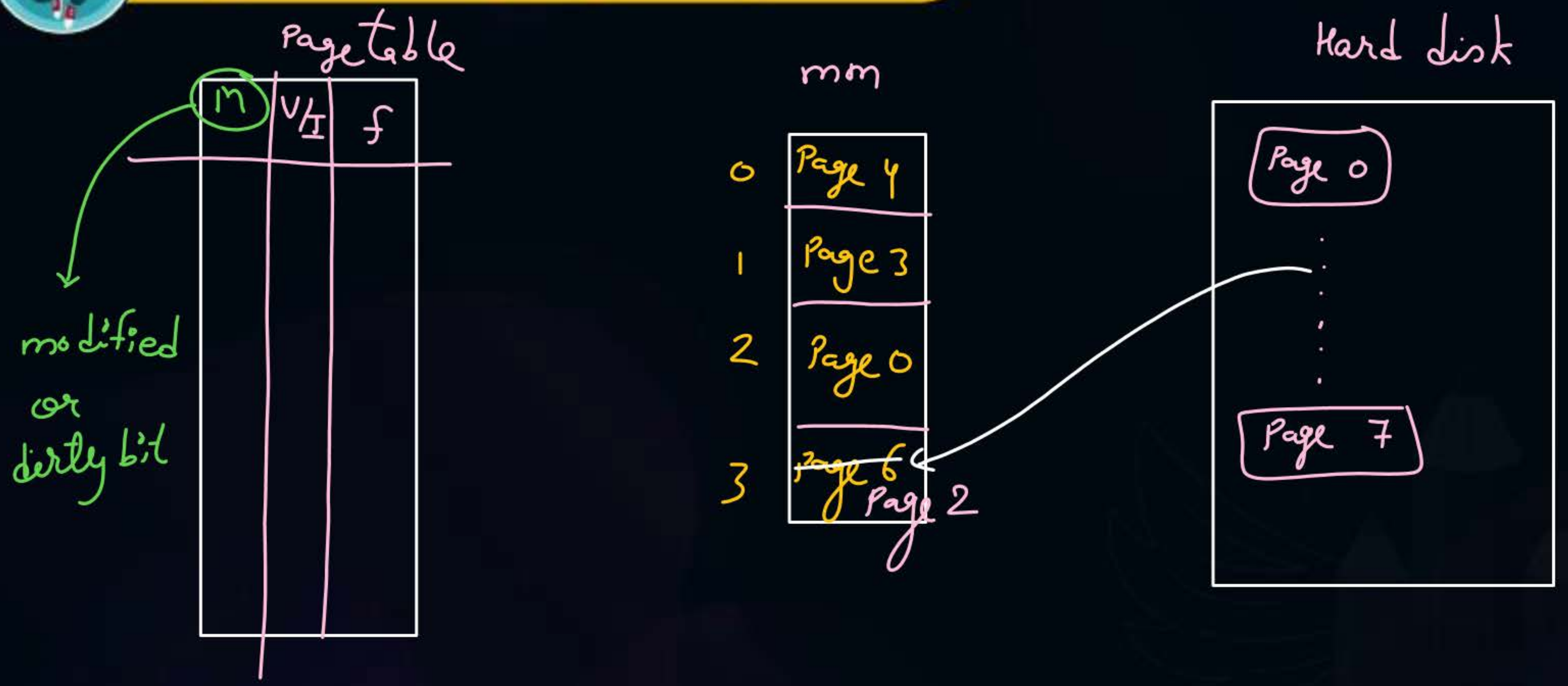
0 or 1

0 → invalid ⇓ page not in mm

1 → valid ⇓ page in mm

	V/I	f
0	1	2
1	0	
2	~~0~~ 1	3
3	1	1
4	1	0
5	0	
6	~~1~~ 0	3
7	0	

Topic : Saving Page Swap Time



M

0 $\Rightarrow$ CPU performed only read in page $\Rightarrow$ if such page is replaced then no write back to disk needed

1 $\Rightarrow$ —— || —— write in page $\Rightarrow$ These will require page write back to disk if replaced

2 mins Summary

Topic Virtual Memory

Topic Page Fault

Happy Learning

THANK - YOU